

Knowledge Graph Implementation Plan

An eight-phase pipeline that transforms the EU cosmetic ingredients database into a taxonomically validated, chemically enriched knowledge graph — offline-first, correction-traceable, and re-runnable from read-only source data.

13,622	3,862	33,638	202
Ingredients	Bio-source rows	2026 Inventory	Families (APG IV)

0	Data Profiling	DONE
Full structural audit of COSING_CAS.csv: encoding quirks, column semantics, sentence patterns in the description field, bio-source row detection, and an inventory of every data-quality issue. KEY TASKS - Catalogue BOM, CRLF, U+00A0 header, CAS leading-whitespace issues - Profile all 9 columns (null rates, cardinality, value distributions) - Classify 3,862 bio-source rows by description sentence pattern (95.4% formulaic) - Identify 15 family misspellings, 29 taxon typos, 115 INCI-vs-description conflicts - Enumerate old family names requiring APG IV normalization (32 mappings) OUTPUTS data-profile.md corrections/*.csv		

1

Data Cleaning & Taxon Parsing

DONE

Load the read-only CSV into DuckDB with format normalization. Parse genus and species via dual-path (INCI tokens + description text). Cross-validate, apply genus-scoped typo corrections, normalize families through typo → APG IV two-stage mapping.

KEY TASKS

- 01_load.py — UTF-8-sig, snake_case columns, CAS splitting, date parsing
- 02_parse_taxon.py — Dual-path parser with synonym stripping, INCI prefix handling
- 03_resolve_conflicts.py — GBIF-based conflict resolution (synonym collapse)
- 04_clean_taxon.py — Family normalization, split-family flagging, final extract table

RESULTS

99.35%	98.1%	236→189	0
Genus resolution	Dual-path agreement	Family dedup	Unresolved conflicts

OUTPUTS

kb.duckdb | cleaning-report.md

2

Taxonomic Hierarchy (GBIF)

DONE

Query the GBIF Species Match API for every parsed binomial to obtain the accepted name and full hierarchy (kingdom → family). Resolve the 67 split-family rows (Liliaceae, Scrophulariaceae, Myoporaceae).

KEY TASKS

- 05_gbif_taxonomy.py — Threaded GBIF lookup with kingdom filtering, HIGHERRANK rejection
- Retry with &kingdom;=Plantae hint for 'Multiple equal matches' responses
- Genus-level fallback for unmatched species; attach hierarchy columns to extract
- Cache all responses in gbif_cache.json with retrieval timestamps

RESULTS

1, 263	3, 179	202
Taxon (2019)	Taxon (2026)	Families final

OUTPUTS

taxonomy.csv | taxonomy_v2026.csv | gbif-taxonomy-report.md

3

Parts & Process Extraction

DONE

Extract plant parts and processing methods from INCI name tokens using controlled vocabularies. Three-pass: INCI tokens → process-implied parts (seedcake→seed) → description regex fallback.

KEY TASKS

- 06_parts.py — Token matching against vocab tables (73 part tokens → 28 normalized)
- Process vocabulary with implies_part column (40 tokens → 28 processes)
- Modifier extraction (17 tokens: expressed, hydrolyzed, rectified, etc.)
- Cell-culture detection (10 tokens: callus, meristem, protoplast, etc.)

OUTPUTS

vocab/plant_part.csv | vocab/process.csv | parts-process-report.md

4a

LOTUS Compound Enrichment

DONE

Join the LOTUS natural products database (674,454 structure–organism–reference triples) to the knowledge graph via GBIF accepted names.

KEY TASKS

- 13_lotus.py — Load gzipped LOTUS CSV, join on accepted binomial
- Build species_compounds_v2026 table (126,775 species–compound pairs)
- Coverage analysis by family, genus, and kingdom

RESULTS

2, 219	126, 775	674, 454
Species w/ compounds	Compound pairs	LOTUS triples

OUTPUTS

species_compounds_v2026 | lotus-report_v2026.md

4b

PubChem CAS→Structure

DONE

Look up molecular structures for every CAS number via PubChem PUG REST API. Second join path to chemical data, especially effective for 9,760 synthetic ingredients with no species link.

KEY TASKS

- 14_pubchem.py — CAS → CID → SMILES/InChIKey/formula (3 workers, 0.75s delay)
- Read-close-fetch-reopen pattern to avoid DuckDB single-writer lock
- Checkpoint every 500 lookups; distinguish 404 from transient HTTP errors

RESULTS

10,600	9,796
CAS cached	Distinct CAS

OUTPUTS

cas_structure_v2026 | pubchem-report_v2026.md

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Regulation Layer (parallel)

DONE

Fetch Annex II–VI substance data from the live CosIng API and the full 2026 inventory (33,638 records). Link regulation into DuckDB. Compare 2019 snapshot against the live database.

KEY TASKS

- 08_fetch_annexes.py — Annex II–VI via paginated API (2,389 entries)
- 10_fetch_inventory.py — Full inventory via lexicographic partitioning
- 09_link_regulation.py — regulation, ingredient_citation, ingredient_cmr tables
- 11_load_inventory.py / 12_normalize_2026.py — Load and schema-map 2026 data

OUTPUTS

regulation (2,389) | inventory_2026 (33,638) | regulation-report.md

5

Mining Hidden Associations

NEXT

Cross-reference the taxonomic hierarchy, chemical compounds, regulation layer, and parts/process data to surface non-obvious relationships.

KEY TASKS

- Gap recommendations — Which families dominate each Function? Which species are absent?
- Regulatory blind-spot analysis — Restriction rate per family; unrestricted congeners
- Convergent-use detection — Same Function across distant families
- Chemotaxonomic consistency — Does the compound profile support the grouping?
- Species × Part comparison — Same species, different parts/processes
- Compound-set similarity — Jaccard similarity, top-k neighbors (k≈5–10)

All conditions met: 2,219 species with compound sets, 376 families, regulation layer, and parts/process data loaded.

6
Vault Note Generation
PENDING

Generate structured Obsidian notes at aggregated levels. Machine-generated sections are fenced with AUTO markers so re-runs never overwrite human notes.

KEY TASKS

- Species notes (~1,500–1,800): frontmatter with accepted name, GBIF key, family
- Family notes (~200): species list, Function distribution, restriction summary
- MOC indexes: Function (70), Part (28), Process (28)
- Dataview-compatible dynamic indexes

7
AI Query Layer
PENDING

Four-tier hybrid retrieval: exact lookup, structured filtering (DuckDB SQL), graph traversal, and vector search (description field only — never for taxonomic names).

KEY TASKS

- Exact lookup — Dictionary: species / INCI / CAS → record ID
- Structured filter — SQL over DuckDB with Function, Annex, family conditions
- Graph traversal — Adjacency-table walks for 'related' queries
- Vector search — Embedding retrieval for fuzzy NL questions (description only)

Do not use vector search for taxonomic names. Systematic naming produces near-identical embeddings; vector retrieval will confidently return the wrong species.

Ontology Summary

Ingredients (all)	13,622
Bio-source extracts	3,862
Species binomials	~1,700
Genera	711
Families (APG IV)	202
Functions	70
Plant parts	28
Processes	28
Regulation entries	2,389
LOTUS compound pairs	126,775

Design Principles

Read-only source. The original CSV is never modified (chmod 444). All corrections live in side-car tables under `_data/corrections/` with full evidence trails.

Re-runnable. `./run_all.sh` rebuilds the entire database from scratch. Human decisions in `conflicts.csv` are preserved across re-runs.

Offline-first. Phases 0–3 require no network. GBIF, LOTUS, and PubChem results are cached locally; the system works fully offline after initial fetch.

No LLM for extraction. INCI is a standardized naming system. Rule-based parsing achieves 99.35% genus resolution. LLM is only used for conflict adjudication with external evidence, never for name extraction.